

Lesson 4

Extensibility

Lesson objectives

- Describe the core Sterling File Gateway application files.
- Explain the procedure to re-brand the user interface.
- Describe the procedure to add custom protocols.
- Explain the procedure to add custom event code.

Introduction

The core Sterling File Gateway application files are:

- Property file
- Business process
- Database
- Mailbox

Property file

The property files help to customize, and diagnose problems pertaining to the operation of Sterling File Gateway. You can maintain customized property file settings with a customer override property file.

Property files are stored in <Install_dir>/properties folder. A few Sterling File Gateway property files are described in the table.

Property File Name	Description
Filegateway_eventcodes.properties	Controls the operation of event codes in Sterling File Gateway.
Filegateway.properties	Controls the operation such as PGP pack and unpack, concurrent session timeout limits during routing, of Sterling File Gateway.
Filegateway_ui.properties	Controls the location of the help systems for Sterling File Gateway, myFileGateway setting.
Filegateway_protocol.properties	Controls the protocols available when specifying the search criteria for FTPIN, FTPOUT, SFTPIN, SFTPOUT.

Business process

A business process is a series of linked software and possibly human activities that accomplishes a business goal. The activities in a business process are primarily performed by components known as services and adapters, and are orchestrated by Sterling B2B Integrator integration engine.

The following table provides the description of a few system business processes used for Sterling File Gateway operations:

Business Process	Description
FileGatewayDeliverFTP	Delivers files to consumers using the FTP protocol.
FileGatewayMailboxRoute	Gathers all mailbox messages available for routing.
FileGatewayMailboxRouteArrivedFile	Routes individual messages identified by FileGatewayMailboxRoute as available for routing.
FileGatewayRoutePGPPackageDocument	Routes PGP packaged documents.
FileGatewayRoutePGPUnpackageDocument	Routes PGP unpackaged documents.
FileGatewaySendMessage	Sends messages from File Gateway.

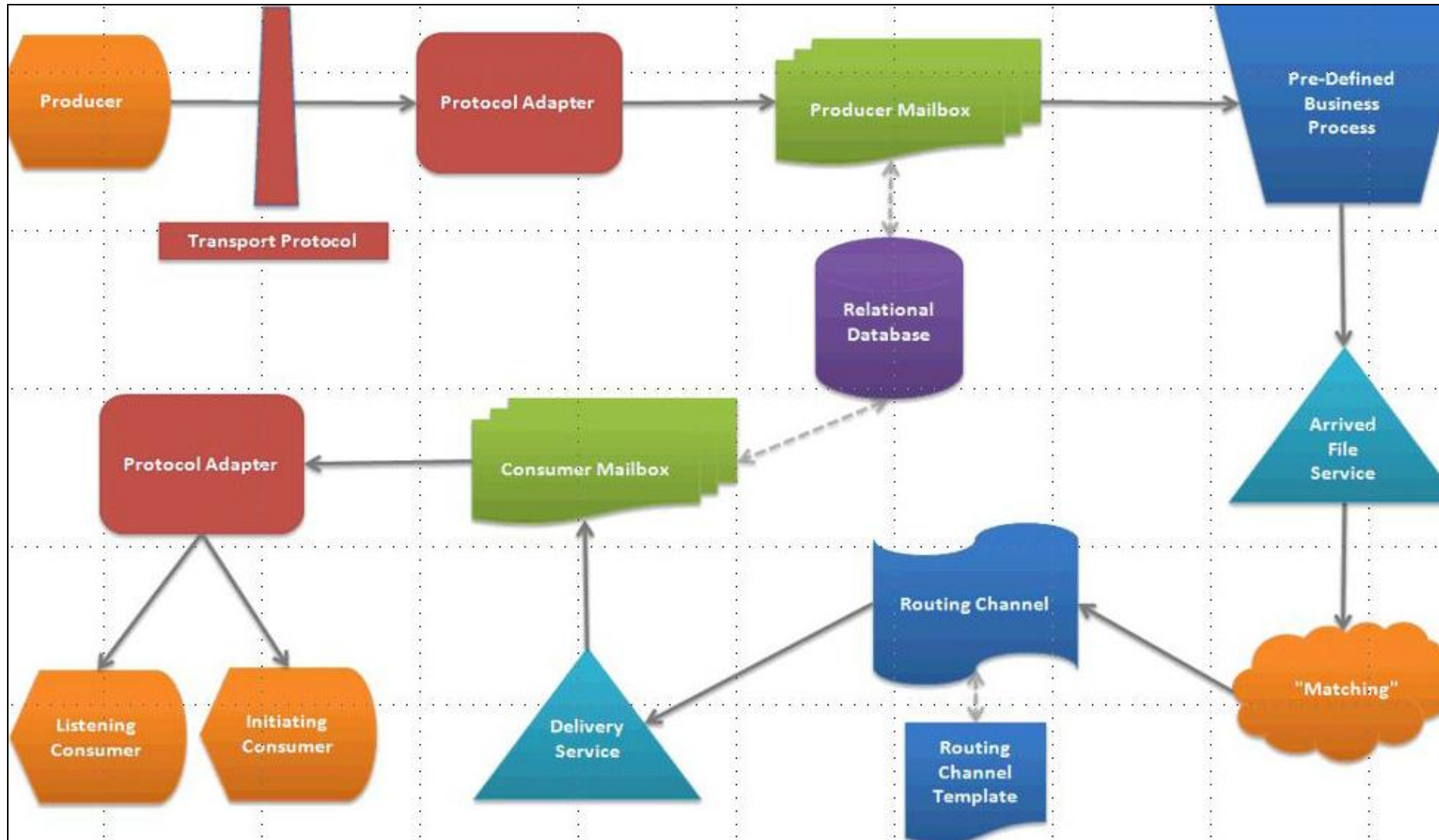
Listening producer API

Listening producer allows Sterling File Gateway to access a user defined server and poll for documents to kick off Sterling File Gateway routes. The current functionality allows for initiating producers, which means you place a file into the producer's mailbox in order to start the transaction.

The list of business processes in Sterling File Gateway 2.2.6.1 for listening producers, mass replay, FTP and SFTP are:

- FileGatewayListeningProducer
- FileGatewayListeningProducerFTP
- FileGatewayListeningProducerSFTP
- FileGatewayProcessFTPReply
- FileGatewayProcessReroute

Sterling File Gateway process flow



Database

There are around 89 built-in tables for Sterling File Gateway application in Sterling B2B Integrator suite.

The screenshot displays the IBM Data Studio Database Administration window for the STERLING database. The left pane shows the Administration Explorer with the database structure tree. The main pane shows the Task Launcher with a search filter 'FG' applied, displaying a list of tables in the DB2INST1 schema.

Schema	Name	Organization	Perce...	Row Count	Log Mode	Primary Key	Partitic
DB2INST1	FG_ARRIVEDFILE	ROW-ORGANI...	0			FG_ARRFILE_PK	
DB2INST1	FG_C_FLR	ROW-ORGANI...	0			FG_C_FLR_PK	
DB2INST1	FG_C_FLR_PRM	ROW-ORGANI...	0			FG_CFLRPRM_...	
DB2INST1	FG_C_FLR_PRM_TYPE	ROW-ORGANI...	0			FG_CFLRPRM...	
DB2INST1	FG_C_FLR_TRANS	ROW-ORGANI...	0			C_FLR_TRANS...	
DB2INST1	FG_C_FLR_TYPE	ROW-ORGANI...	0			FG_C_FLR_TYP...	
DB2INST1	FG_C_FST	ROW-ORGANI...	0			FG_C_FST_PK	
DB2INST1	FG_DELIVCHAN	ROW-ORGANI...	0			FG_DELIVCHA...	
DB2INST1	FG_DELIVCHAN_TMPL	ROW-ORGANI...	0			FG_DC_TMPL_...	
DB2INST1	FG_DELIVERY	ROW-ORGANI...	0			FG_DELIVERY_...	
DB2INST1	FG_EVENT	ROW-ORGANI...	0			FG_EVT_PK	
DB2INST1	FG_EVENTATTR	ROW-ORGANI...	0			FG_EVTATTR_...	
DB2INST1	FG_EVENT_ATTR	ROW-ORGANI...	0			FG_RT_EVT_A...	
DB2INST1	FG_EVENT_DEF	ROW-ORGANI...	0			FG_REV_DEF_PK	

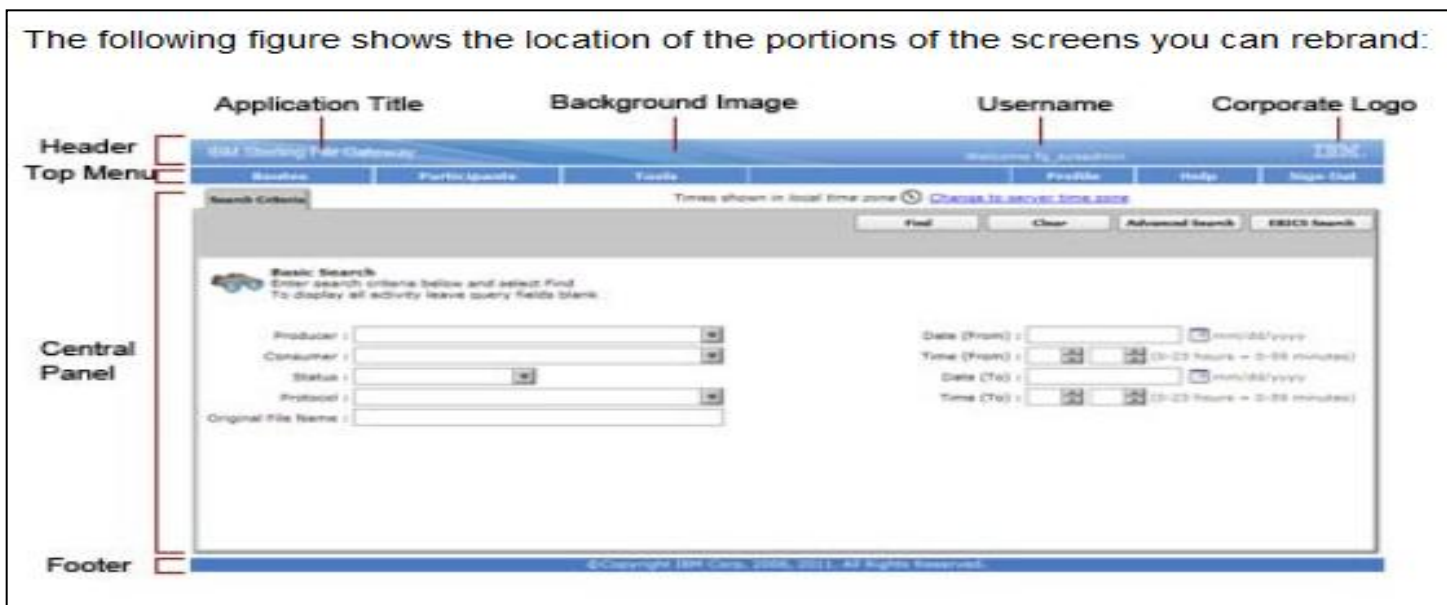
Connection : 192.168.40.100 - db2inst1 - STERLING

Showing 39 of 895 items

Rebranding

Sterling File Gateway interfaces can be customized to re-brand the product name, menus of Sterling File Gateway, and myFileGateway user interface to incorporate new company logo or other special content.

The screen shot represents a few standard names such as application title, background image, top menu, logo to customize the user interface of Sterling File Gateway.



Custom Protocol

Sterling File Gateway supports to customize protocols for listening consumers.

The customizing protocol procedure determines the text that is shown in the Partner management wizard. Customized protocol, allows you to select, as new choices in the Sterling File Gateway interface:

- when creating a community.
- when creating a listening consumer in a community.

The screenshot shows a dialog box titled "Add Community: Protocol". Inside the dialog, there is a section titled "Make Protocols Available to Partners" with a blue header. Below this header, there are five checkboxes. The first two are checked: "Partner Initiates Protocol Connections to Mailbox" and "Partner Listens for Protocol Connections". The next three are unchecked: "FTP or FTPS", "Sterling Connect:Direct", and "SSH/SFTP". At the bottom of the dialog, there are three buttons: "< Back", "> Next", and "Cancel". A mouse cursor is pointing at the "FTP or FTPS" checkbox.

Creating Custom Protocol

To Add Custom Protocols

- 1) Write a business process that implements the custom protocol.
- 2) Use the existing *AFTEExtensions.xml* as a guide to create an *AFTEExtensionsCustomer.xml* file to describe the protocol which is present under
- directory: <install_dir>\container\Applications\aft\WEB-INF\classes\resources\xml.
- 3) Use the existing *AFTEExtensions.properties* file as a guide to create the *AFTEExtensionsCustomer.properties* file which is located under
<install_dir>\container\Applications\aft\WEB-INF\classes\resources

System event codes

Event Code	Name	Description	Default Subscription Settings
FG_0050	MailboxAuthorizationFailure	When a user fails to access a mailbox that is monitored	Operators
FG_0201	BusinessProcessStarting	When a business process is started, with the reason for starting it stored in context.	Only in activity details.
FG_0202	BusinessProcessSuccess	When a business process is successfully completed, with a link to the business process.	Only in activity details.
FG_0203	SendMessageBusinessProcessStarted	When a business process is successfully started, with a link to the business process.	Only in activity details.
FG_0250	BusinessProcessFailure	When a business process fails, with a link to the business process	Operators
FG_0401	FactDiscovery	When a fact is discovered by the route, for example, matching a pattern on the file name.	Only in activity details.
FG_0404	ProducerFound	When a producer and user are identified based on the arrived file.	Only in activity details.
FG_0407	ConsumerFound	When a consumer is identified based on the route plan.	Only in activity details.
FG_0408	ArrivedFileStarted	When a file arrives to be routed	Operators, producers, and consumers
FG_0409	ArrivedFileDeterminingRoutes	When an arrived file starts determining routes.	Only in activity details.
FG_0410	ArrivedFileRouting	When an arrived file starts processing routes.	Only in activity details.
FG_0411	ArrivedFileRouted	When an arrived file completes all its routes	Operators, producers, and consumers
FG_0412	RouteStarted	When a route is started from an arrived file.	Only in activity details.
FG_0413	RouteDeterminingDeliveries	When a route starts determining deliveries.	Only in activity details.
FG_0414	RouteDelivering	When a route starts processing deliveries.	Only in activity details.
FG_0415	RouteComplete	When a route completes all its deliveries	Operators, producers, and consumers
FG_0416	DeliveryStarted	When a delivery is started from a route.	Only in activity details.
FG_0417	DeliveryTransforming	When a delivery starts to transform the consumer file.	Only in activity details.

Event Code	Category
00	Producer File Transfer
02	Routing Business Process
04	File Gateway General
05	Route Plan Determination
06	Route Plan Determination
07	Consumer File Transfer

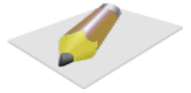
Custom Event Codes

- Each activity during the progress of a file transfer generates an event code. These codes are displayed in the activity details to enable Partners and Operators to see the progress and navigate to more details.
- System Administrators can create custom event codes and modify the permissions for standard event codes.
- Some event codes trigger notifications to subscribers. System Administrators can change the permissions for an event code by using the `customer_overrides.properties` file to override values in the `filegateway_eventcodes.properties` file.

Custom Event Codes

- Event Code: The event code in the template you created is CUST_0265. It consists of keyword underscore 4-digit code. It is used to identify all the attributes for a particular event code.
- Keyword - any text, optimally about four characters long, but cannot be FG nor AFT
- 4-digit Code - consists of two 2-digit codes - a category identifier and a unique identifier. No two events can have the same 4-digit code.
- Category Identifier - The first 2- digit code is a category identifier.
- Unique Identifier - The second 2-digit code is the unique identifier for the event within the category. 00 - 49 are successful events and 50 – 99 are error events. Error events show up as red in Sterling File Gateway.

Exercise



1. Re-brand Sterling File Gateway.
2. Adding customer protocols.

Completed lesson objectives

- Describe the core Sterling File Gateway application files.
- Explain the procedure to re-brand the user interface.
- Describe the procedure to add custom protocols.
- Explain the procedure to add custom event code.

Questions



Completed unit objectives

- Explain the Sterling File Gateway dynamic route.
- Discuss the Sterling File Gateway file layers.
- Generate Sterling File Gateway reports.
- Explain the extensibility of Sterling File Gateway.